

Framing Financial Feedback: Narrative vs. Factual AI Explanation in an Educational Simulation

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This paper presents a study of how narrative versus factual AI-generated feedback shapes college students' comprehension of financial literacy concepts within an interactive "what-if" decision simulation. Twenty participants used one of two versions of the system, with each delivering identical underlying information through a different explanation style. While no significant difference was observed in the quantitative learning gains between the two conditions, a qualitative pattern emerged, in which participants in the factual condition were less likely to characterize financial concepts as "common sense". This suggests that explanation style shapes learners' epistemic positioning toward the content, such as being aware of financial complexity. We discuss what these findings suggest for technical communicators and designers working with AI-generated feedback in educational tools.

CCS CONCEPTS • Human-centered computing → Human computer interaction (HCI) → HCI design and evaluation methods; Applied computing → Education → Computer-assisted instruction

Additional Keywords and Phrases: communication design, AI-mediated communication, explainable AI, financial literacy, serious games

1 INTRODUCTION

Students with limited financial literacy often face barriers to accessing practical, relatable, and engaging financial education. Traditional financial education programs frequently rely on abstract concepts, static lessons, or one-size-fits-all curricula that fail to address individual learning needs and adapt to learners' lived experiences [1,2]. As a result, students may struggle to connect theoretical knowledge to the complex, dynamic decisions they encounter in daily life, such as budgeting, saving, and evaluating trade-offs between spending and investing. AI-mediated learning systems open new flexibility in how feedback is communicated to learners, but the form that feedback takes is itself a communication design decision: the same underlying outcome can be delivered through very different explanation styles, and the implications of choosing one over another remain underexplored.

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To address this gap, this paper introduces a “what-if” decision-making game that integrates gamification and explainable AI (XAI) to create a more interactive and reflective financial learning environment. Players explore the consequences of financial decisions in simulated real-world scenarios, allowing them to experiment, observe outcomes, and adjust strategies without real-world risk, while the AI component provides immediate guidance and feedback on each choice. This research raises the central question: *how does narrative versus factual framing in AI-generated feedback affect how college students comprehend and internalize financial literacy concepts, and what does that suggest about how explanation form shapes learners’ relationship to the content?*

2 RELATED WORKS

2.1 Communication design and learning

Mehlenbacher [3] proposes a sociocognitive framework for communication design and argues that audiences should be understood as active learners navigating complex, ill-structured information spaces. Individual comprehension is shaped by internal cognitive processing and environmental and cultural context, including how information and feedback are framed. Design features that affect framing can either facilitate or hinder learners’ deeper understanding. In the context of financial literacy, Mehlenbacher’s framework implies that how a financial concept is explained should matter as much as what is explained. His work, however, predates AI-generated feedback systems, leaving open the question of whether different AI explanation styles, narrative versus factual, produce meaningfully different learning outcomes.

2.2 Financial literacy, gamification, and AI in education

Prior work demonstrates that serious games can produce measurable gains in domain knowledge and engagement when they embed experiential, consequence-driven mechanics. Hu et al. developed *Financial Decision Buddy* [1], a decision-support tool that integrates machine learning with a Maximum Expected Utility framework to help young adults make more informed financial decisions through real-time AI guidance. *DreamScape* [2] is a narrative-driven serious game in which players navigate financial dilemmas through interactive storytelling and branching dialogues. Players make savings and budgeting decisions that directly shape the story’s progression, and early results are promising in introducing key financial concepts and engaging users. *Cash Action* [4] teaches financial concepts through collaborative, consequence-driven multiplayer gameplay. More recent work has begun integrating large language models directly into financial education, as in Chen’s study of LLM-supported financial digital literacy [5]. These studies show that interactive, AI-driven tools can support learning and empowerment in financial decision-making, but how AI explains and contextualizes outcomes has received less attention than the surrounding game mechanics.

2.3 Explainable AI and explanation style

Martens et al. [6] introduce *XAIstories*, a framework that converts technical model outputs into narrative stories using large language models, and find that narrative framing significantly improves comprehension, persuasiveness, and perceived trust over feature-based explanations. Celar and Byrne [7] find that domain familiarity significantly shapes how users engage with AI explanations. Users with prior knowledge are better able to evaluate and critically challenge the explanations they receive, which suggests that explanation styles need to be calibrated to the learner’s existing knowledge to be effective. Aryal and Keane [8] survey the broader landscape of semi-factual explanations in XAI, mapping how distinct counterfactual, semi-factual, and causal explanation forms are defined and evaluated as different

strategies for communicating AI decisions. These studies establish that explanation form matters for how users engage with AI output, but they have not addressed how such forms function in serious games aimed at non-expert learners.

3 SYSTEM DESIGN AND METHODOLOGY

The system is an interactive "what-if" financial decision-making simulation in which players navigate budgeting, saving, and borrowing scenarios across three levels and receive immediate AI-generated feedback on their choices. The central design variable is explanation style. Both conditions convey the same underlying information, and only the communication strategy differs.

3.1 Explanation conditions

Both conditions reinforce the same underlying financial principles, present each choice as part of the player's broader financial journey, and show the cause-and-effect relationships between decisions and outcomes. The only difference between them is the surface language. The factual condition delivers concise, data-driven feedback using objective metrics such as account balances, interest calculations, and budgeting percentages, with technical financial terminology used throughout. The narrative condition delivers the same feedback as sequential, story-like reflections in plain, everyday language, without relying on technical financial vocabulary.

3.2 GPT customization and configuration

Two versions of GPT were customized on the OpenAI ChatGPT platform and built on the GPT-4 model, each corresponding to one of the two explanation modes. Both were customized to (1) align with a set of predefined scenarios covering budgeting, saving, and borrowing, (2) maintain consistency in the information conveyed and ensure that the only difference between modes was the explanation style, and (3) respond in ways that are clear and relevant for college students with varying levels of financial literacy. To eliminate live-prompt variance during data collection, all participant-facing scenarios and explanations were finalized in advance and delivered through a high-fidelity, fully interactive Figma prototype designed as a visual novel-style interface, ensuring identical content delivery within each condition.

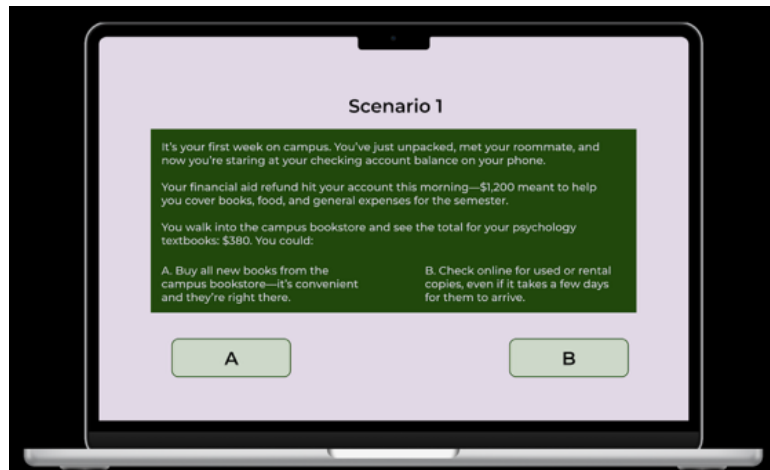


Figure 1: Figma prototypes of one scenario and its answer choices in the Narrative mode on a desktop view.

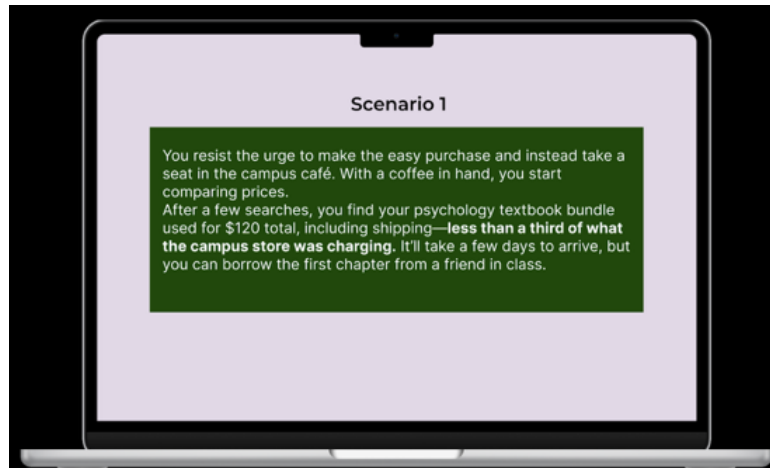


Figure 2: Figma prototypes of one scenario and its explanation in the Narrative mode on a desktop view.

3.3 Study design

Twenty college students were randomly assigned to one of the two conditions in a quasi-experimental design. Data collection was structured around three phases, which were pre-, mid-, and post-study surveys. The pre- and post-study surveys tracked the acquisition and retention of key financial principles quantitatively, while the mid-study tracked the comprehension qualitatively. The user study was followed by a brief verbal debrief asking whether the scenarios made sense, whether the concepts felt like common sense, and whether participants would have known the concepts without being told.

4 RESULTS

The most revealing differences between conditions emerged qualitatively. Participants in the factual condition were less likely than those in the narrative condition to characterize financial concepts as "common sense" during the verbal debrief, which suggests that data-centered feedback may make financial concepts feel less self-evident, even on first encounter. Because both conditions delivered identical information about the same scenarios with the same computed consequences, the explanation's rhetorical form is the most likely reason for this difference. Other contributing factors may play a role, such as individual differences between the small groups. Even so, because the controlled design holds scenario content, sequence, and outcomes constant across conditions, explanation style is a likely contributor. On this evidence, explanation style appears to do more than transmit information: it may shape how learners position themselves relative to the content and what they perceive as new knowledge.

Thematic analysis of the mid-study comprehension surveys revealed recurring themes across both conditions, including delayed gratification, budgeting and financial prioritization, debt avoidance, and interest rate mechanics, with delayed gratification emerging consistently across all three levels. This suggests that the simulation's experiential structure supported conceptual reflection regardless of explanation style.

Quantitative analysis showed no statistically significant differences in objective knowledge gains between conditions. The sample size was small ($N=20$), and participants were drawn from a single private liberal arts institution whose selective admissions and high cost of attendance may be associated with higher baseline financial exposure or

confidence, potentially producing a ceiling effect against which short-term differences would be difficult to detect. These limitations bound the present claims but do not weaken the qualitative observation, which holds within-study and reveals a hypothesis worth testing further.

5 FUTURE WORK

Future work will expand the simulation into a fully deployed web application with interactive features including a live financial dashboard, a conversational AI advisor, and a what-if scenario toggle that lets learners explore counterfactual outcomes in real time. The study design will be expanded in several aspects: broader and more diverse recruitment to mitigate the institutional ceiling effect observed here, and individual differences as moderating variables, specifically Need for Cognition [9], which captures how much learners enjoy effortful thinking, and the Transportation Scale [10], which measures narrative engagement and immersion. Retention will also be assessed after a one-week delay to evaluate the long-term effects of explanation style on financial literacy. Together, these additions aim to test whether the qualitative effect observed here is generalizable and mediated by individual differences. This would also build toward a more complete account of how AI explanation design interacts with learner cognition and communication outcomes in financial literacy education.

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